

Sustainable Natural Farming System

Paedar Qudratti Nizam Kashatqari (PQNK)

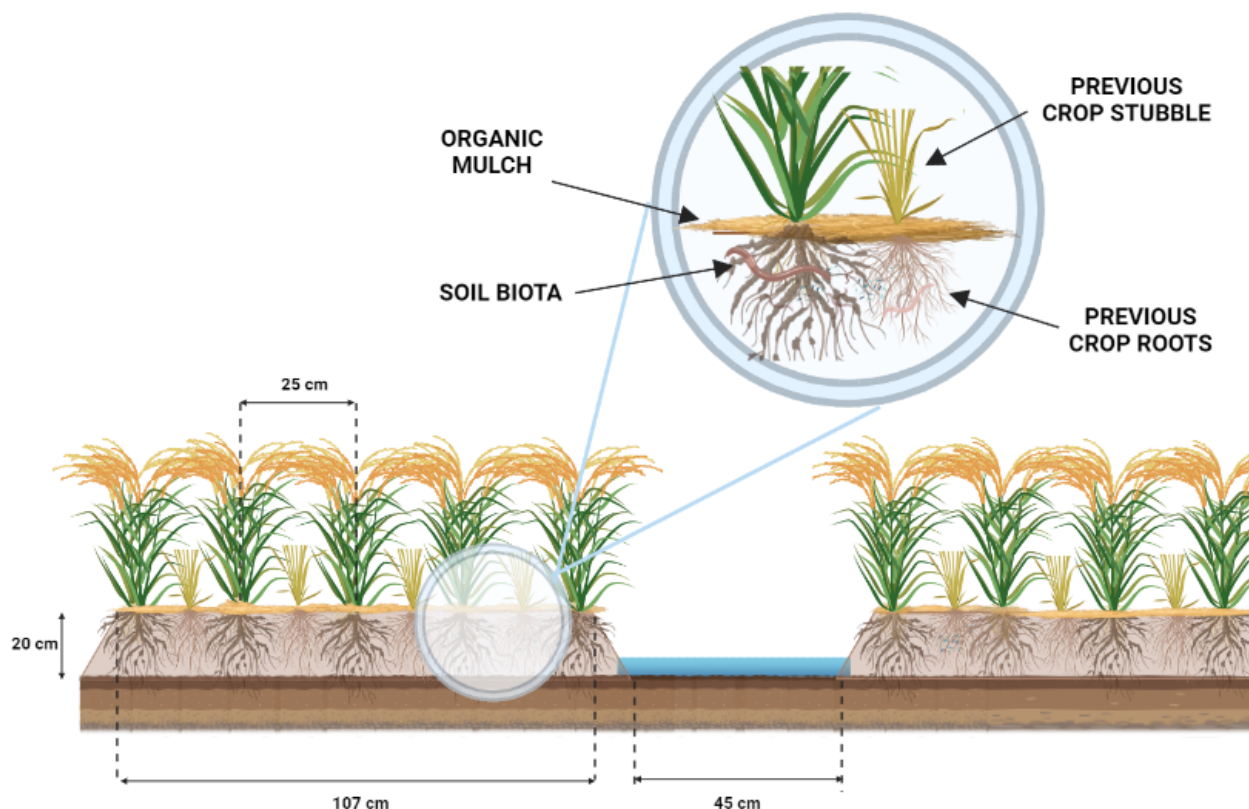
What is PQNK

PQNK is the abbreviation of *Paedar Qudratti Nizam Kashatqari* (to be pronounced as picnic) which means sustainable natural farming system in Urdu and it is also referred to as Paradoxical Agriculture. Asif Sharif has conceptualised PQNK through first-hand experience on his own land in the Indo-Gangetic Plains of Pakistan. He later founded Pedaver, a company that works with thousands of farmers to disseminate the PQNK system and develop appropriate mechanisation. Initially, after identifying that inundation, soil disturbance, and bare land are contrary to the natural processes of soil fertility and vegetation, Mr Asif developed the raised bed planting system. Rice was the first crop tested on Asif's raised beds as conventionally it is grown under flooded conditions. This was to demonstrate that water serves as a nutrient carrier, and only the water absorbed by the roots and transpired from the leaves is needed for the plant development. At this stage, Asif came across the System of Rice Intensification (SRI) and gained insights into optimal plant population, i.e. row spacing and plant-to-plant distance. After achieving success by reducing rice plant density, the next objective was to maintain soil coverage. Rice crop residues had been utilised for this purpose. Soon PQNK developed into a more comprehensive farming method for all crops and trees that combines SRI and Conservation Agriculture (CA), with natural farming as an overarching principle. The PQNK approach replicates natural dynamics in agricultural production systems, like avoiding the use of synthetic amendments, keeping upland for vegetation and lowland for water supply and drainage of excessive water, as shown below.



PQNK Agricultural Practices

Under PQNK, crops are cultivated on permanent no-till raised beds that are formed on levelled, non-compacted land. For this, the beds' size is adapted to the size of the tractor and machinery wheels, so that the wheels are driving through the furrows, hence not compacting the beds. Beds are covered with organic mulch that can be locally procured by leaving crop residues on the field after harvesting and by using cover crops in between cash crops. Irrigation is applied in the furrows between beds. Nutrient uptake from the soil is optimised by plant spacing and hardpan breaking. The plant spacing also allows each plant to develop large canopies for catching sunlight and CO_2 from the atmosphere. The plant spacing is to be adapted to the specific requirement of each crop. Under PQNK, crop establishment can be done by both transplanting or direct seeding. Direct planting is preferable, but if choosing to transplant the seedlings, one should do this early and carefully to bring minimal disturbance to the root systems. Intercropping, relay cropping, and under-sowing are techniques that can be used by farmers practising PQNK to promote crop diversification within the farming system. Introducing agroforestry practices such as alley cropping, where trees are placed within the agricultural field, is also a way to increase the available in-field biomass for mulching and to promote biodiversity while increasing farm production.



Why raised beds?

Growing crops on raised beds minimises soil disturbance to protect soil biota, improves soil structure, and promotes carbon sequestration. Also, with machines developed to the size of the raised beds, PQNK facilitates the large-scale implementation of no-till. In PQNK, soil compaction is completely avoided by letting the tires of the heavy machinery drive in the furrows between the beds. This guarantees driving through the land without compromising the soil structure where crops are planted.

Designing the field with raised beds allows for furrow irrigation which improves water-use efficiency. For rice cultivation, which is traditionally implemented in flooded conditions, raised beds facilitate the implementation of SRI principles as they are never submerged. The bed topsoil is kept in aerobic condition, avoiding the degradation of plant roots, while also supporting more abundant beneficial soil organisms. This is particularly useful in areas prone to flooding as raised beds allow proper drainage of water, therefore, protecting crops from suffocating.

Forming raised beds initially requires a significant expenditure of labour or capital (for mechanised construction), but this can lead to lower requirements for labour subsequently. Under PQNK, the formation of raised beds cuts the number of man-hours needed per hectare from 85 to 25, a 70% reduction and a reduction of water usage compared to flooding of 90%.



Why organic mulch?

In a PQNK system, the layer of organic mulch should be thick enough to cover the soil surface and prevent sunlight from reaching the soil. This inhibits weed germination, resulting in a non-chemical strategy for weed management. Maintaining permanent biomass cover on the soil surface also protects the land from overheating in direct sunlight, which adversely affects the soil biota. In fact, temperatures higher than 40°C kill almost all life in the soil. Organic Mulch also buffers the force of winds and storms that erode the topsoil. Further, organic mulch applied as soil cover adds carbon and minerals to the soil system and supports the proliferation of soil microorganisms and mesofauna. The evaporative losses of moisture are also reduced by biomass cover, ultimately improving the water-use efficiency. In addition to this, the retention of non-harvested crop biomass, rootstocks, and stubble in the field after harvesting conserves the stock of carbon in the soil. These materials, when they decompose, support the soil biota and improve soil structure. PQNK makes crop residues a valuable source of biomass as well as a tool for integrated weed and nutrient management, giving farmers incentive not to burn them and thus also improving air quality.



Why optimise plant spacing?

Increasing the spacing between plants, i.e., reducing plant density per m², should be done when soil is managed according to the principles of CA as in PQNK systems. This is because it is natural for tillering crops to grow more profusely in the more fertile soil environment that is created by PQNK practices: more soil organic matter, more soil moisture, more biotic activity, and better soil structure. By allowing each plant to fully develop its root systems and its canopy, more nutrients are absorbed from the soil (and more carbon is stored in the soil), while the photosynthetic process is boosted. The result is an increase in plant productivity which comes together with a reduced expense for seeds in case farmers need to buy them.



Why diversification?

Promoting diversification within the farming system increases biodiversity and results in greater land use efficiency and higher overall farm yield while enhancing carbon sequestration, especially when trees are included. Within a PQNK system, it is important to consider multiple strategies for achieving greater crop biodiversity. Introducing cover crops between seasons or using crop associations such as intercropping, alley cropping, relay cropping, and under-sowing are valuable options. As raised beds need to be permanently covered with organic mulch, diversification strategies can provide the needed amount of biomass to keep the soil covered all throughout the year. Crop diversification also reduces farmers' economic vulnerability and the risks connected to crop failures.



Mechanisation

Pedaver has developed machinery for mechanically forming raised beds and furrows and for mechanised direct planting through heavy mulch with precise plant spacing. Transplanting of young single seedlings in permanent no-till raised beds with precise spacing is also possible thanks to



Pedaver's innovations. Recent machinery even allows for forming the raised permanent beds while direct seeding in one single passage of the tractor. These innovative machines allow the upscale of PQNK systems and the implementation of this system on large-scale farms. The field design with permanent raised beds and furrows for irrigation complies with the utilisation of tractors whose wheels pass on the furrows without disturbing and compacting the cultivated land. In this way, furrows get compacted, resulting in reduced water percolation during irrigation and increased water infiltration in the cultivated beds, ultimately enhancing water-use efficiency. Appropriate mechanisation highly reduces the drudgeries of farm works and facilitates the timely implementation of agricultural activities. Pedaver's innovations focus mainly on the two most labour-intensive activities in the PQNK system: forming the no-till raised beds (a one-time operation) and (trans)planting of crops. In fact, other operations, such as tilling the field and weeding, and spraying of agrochemicals are not needed under the PQNK approach because of the presence of permanent no-till raised beds and heavy mulch.



Results

The PQNK system has been adopted by tens of thousands of Pakistani farmers because of the economic benefits it brings. Crops' performances improve under PQNK because of the better soil environment created by the system. Root systems and canopies get bigger thanks to the optimised plant spacing and breaking the hardpan also gives space for the roots to expand. It all adds up to an increased availability of water and nutrients in the soil. Plants with improved phenotypic traits are not only more productive but also more resistant and resilient against the effect of extreme weather events. PQNK practices result in a natural plant protection and nutrition system that protects the plants against the attacks of pests and diseases and optimises the food uptake of the plants. This lowers the risks for farmers while also reducing the costs related to the use of agrochemicals that negatively affect life in the soil. Also, the reduced amount of labour connected to the adoption of permanent no-till mulched raised beds further cut production costs. Lower costs and better yield in quantity and quality result in greater profit for farmers, as reported in the table below, which is what makes PQNK so attractive among Pakistani farmers.

The positive environmental externalities have great importance in terms of resource conservation and ecosystem restoration. By being completely organic and by avoiding tillage, PQNK systems minimise water and soil contamination while enhancing carbon sequestration and soil biological activities. Moreover, by breaking the hardpan, allowing the rain and pumped-up irrigation water to seep back into the soil, the aquifers are replenished. Also, by avoiding flooded conditions in rice cultivation, PQNK systems highly reduce methane emissions.



Cost-effectiveness of PQNK vs Conventional Agriculture - major crops

	Wheat		Sugarcane		Corn		Potatoes		Rice		Cotton	
Cost heads (Rs/acre)	Conv.	PQNK	Conv.	PQNK	Conv.	PQNK	Conv.	PQNK	Conv.	PQNK	Conv.	PQNK
Land preparation	8500	2000	8500	2000	8500	2000	16000	2000	16000	2000	8500	2000
Seed	7500	1875	10000	750	20000	20000	60000	80000	1500	525	2000	2000
Sowing/planting	4000	4000	8000	4000	4000	8000	8000	6000	12000	4000	4000	4000
Fertiliser	16300	/	25800	/	25800	5000	25800	2000	16300	2000	16300	/
Pesticides	/	/	2000	/	2000	1000	5000	/	2000	/	8000	/
Weedicides/hoeing	2000	/	2000	/	2000	/	2000	/	2000	2000	2000	/
Fungicides	/	/	/	/	/	/	4000	/	2000	/	/	/
Harvesting exp.	8000	8000	32000	40000	8000	8000	10000	10000	8000	8000	8000	8000
Water exp.	8000	2000	12000	2000	10000	2500	5000	2000	12000	2000	8000	2000
Labour/supervision	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000	10000
Land rent	75000	75000	150000	150000	50000	50000	50000	50000	50000	75000	75000	75000
Total cost of production	139300	102875	260300	208750	140300	106500	195800	162000	131800	105525	141800	103000
Yield (bags/acre)	35	40	800	1000	80	80	80	80	40	40	25	30
Production cost per bag	3980	2571.9	325.4	208.7	1753.7	1331.2	2447.5	2025	3295	2638.1	5672	3433.3
Sale price per bag	4400	4400	350	350	2600	2600	3000	3000	5800	5800	7200	7200
Profit per bag	420	1828	24.6	141.2	846.2	1268.7	552.5	975	2505	3161.9	1528	3766.7
Profit per acre	14700	73125	19700	141250	67700	101500	44200	78000	100200	126474.8	38200	113000.1
Profit variation (%)	NA	397.4	NA	617	NA	49.9	NA	76.5	NA	26.2	NA	195.8

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